MIL-M-28787/111A(EC)
10 December 1981
SUPERSEDING
MIL-M-28787/111(NAVY)
18 January 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TERMINATOR, INTERFACE, DIGITAL, KEY CODE MDN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface terminator module.

1.2 Classification This specification covers one class of modules.

Class I as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.7 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MDN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 56 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/111-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrie

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

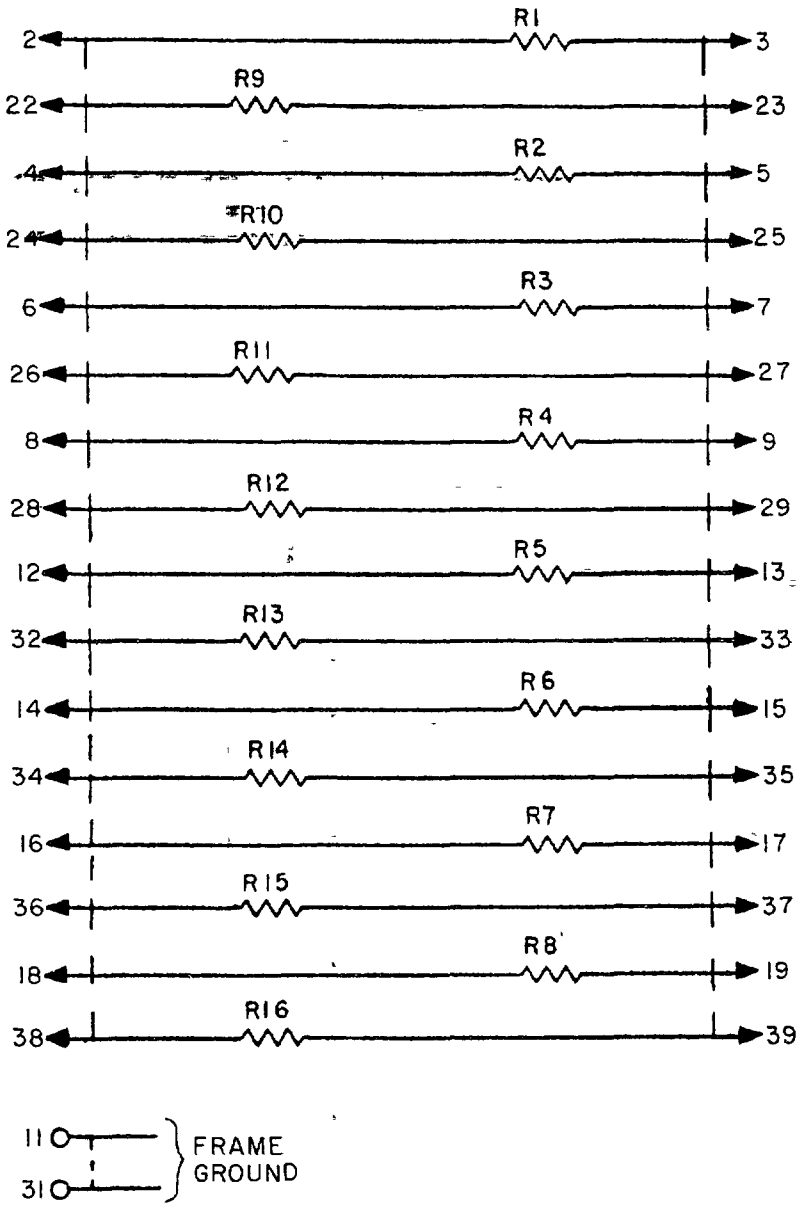
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES 1. R1 - R16 = Sixteen noninductive type, isolated resistor elements.

2. Circuit pins not used = 1, 10, 20, 21, 30 and 40.

FIGURE 1. MDN module function diagram, interface terminator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A10	Module integrity	II	X	10	∞	10	∞	10	∞	10	∞	MΩ
B1	Continuity	III	X	0	2	0	2	0	2	0	2	Ω
B2	Continuity	III	S	0	1	0	1	0	1	0	1	Ω
C1-C16	Resistance	IV	X	73	77	70	80	73	77	70	80	Ω
D1-D16	Capacitance	V	S	0	30	0	30	0	30	0	30	pF
E1-E16	Overshoot	VI	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E17-E32	Undershoot	VI	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
F1-F16	Resistance surge	VII	S	73	77	70	80	73	77	70	80	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent to two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through F shall be performed in any order with the following exceptions. If test conditions A, B, C, or F are part of a sequence of tests, the order of testing shall be as follows A, B, C, F where applicable, followed by the remaining tests in the sequence.

4 5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4 5 1 Operating temperature tests The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through C in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through C shall be performed in the order specified in 4 4.2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4 5 3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A, B, and C shall be performed in the order specified in 4 4 2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests A through F shall be performed in the order specified in 4 4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331.

6 2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3 8)

6 3 Cross reference of classification Military Specification MIL-M-28787/111 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 70001-2658834C dated 19 May 1970 Part number M28787/111-1 supersedes part numbers 2658834 and M28787/111. These parts are interchangeable

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Power dissipation

Maximum	1 7 W
Operating	1 4 W
Quiescent	0 0 W

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6.5 Module function. The module consists of sixteen 75-ohm resistors. Each resistor shall be capable of dissipating 125 milliwatts maximum. The total resistor power dissipation shall not exceed the maximum module power dissipation.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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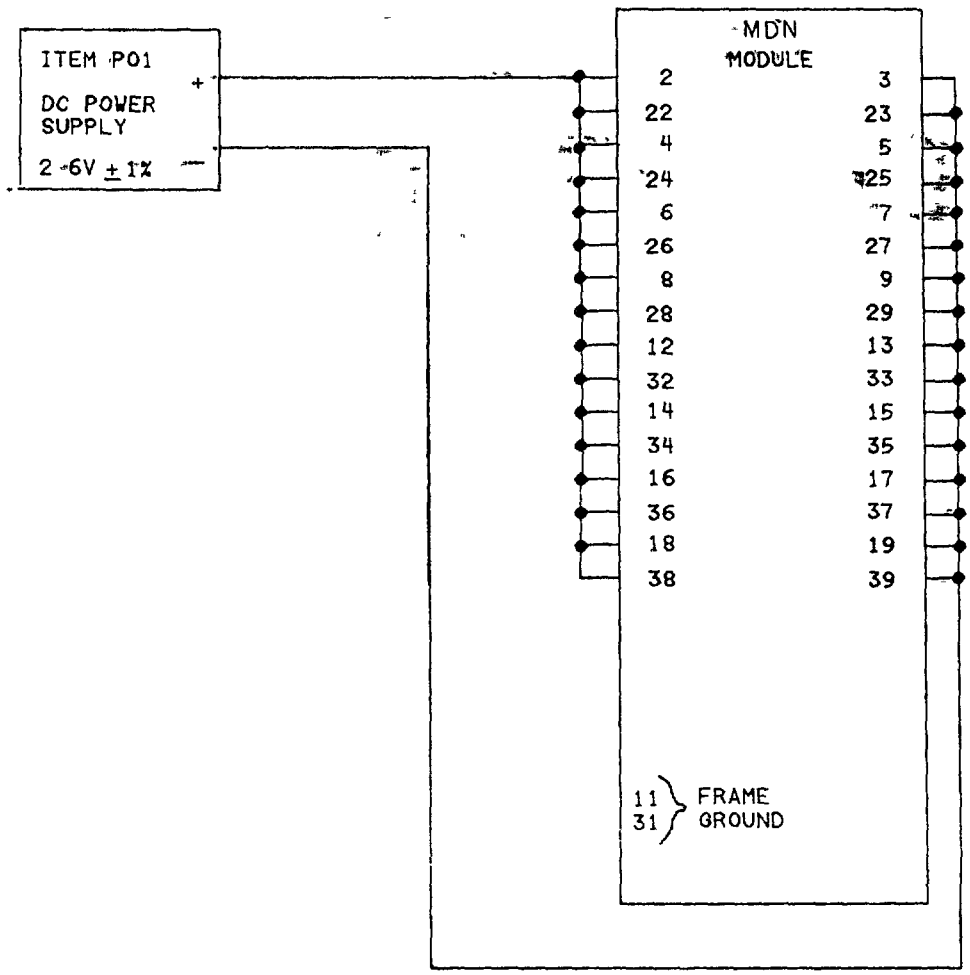


FIGURE 2 Operating temperature, vibration, and life test circuit

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TABLE II. Module integrity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Module integrity	11, 31	All other pins	10	∞	MΩ	3
A2	↓	1,10,20,21,30 40	All other pins	↓	↓	↓	↓
A3		2,4,6,8,12,14, 16,18	23,25,27,29, 33,35,37,39				
A4		2,4,6,8,23,25, 27,29	12,14,16,18, 33,35,37,39				
A5		2,4,23,25	6,8,27,29				
A6		2,23	4,25				
A7		6,27	8,29				
A8		12,14,33,35	16,18,37,39				
A9		12,33	14,35				
A10	↓	16,37	18,39	↓	↓	↓	↓

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TABLE III. Continuity test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
B1	Continuity	11	31	0	2	Ω	3
B2	Continuity	11	Frame	0	1	Ω	3

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TABLE IV. Resistance test.

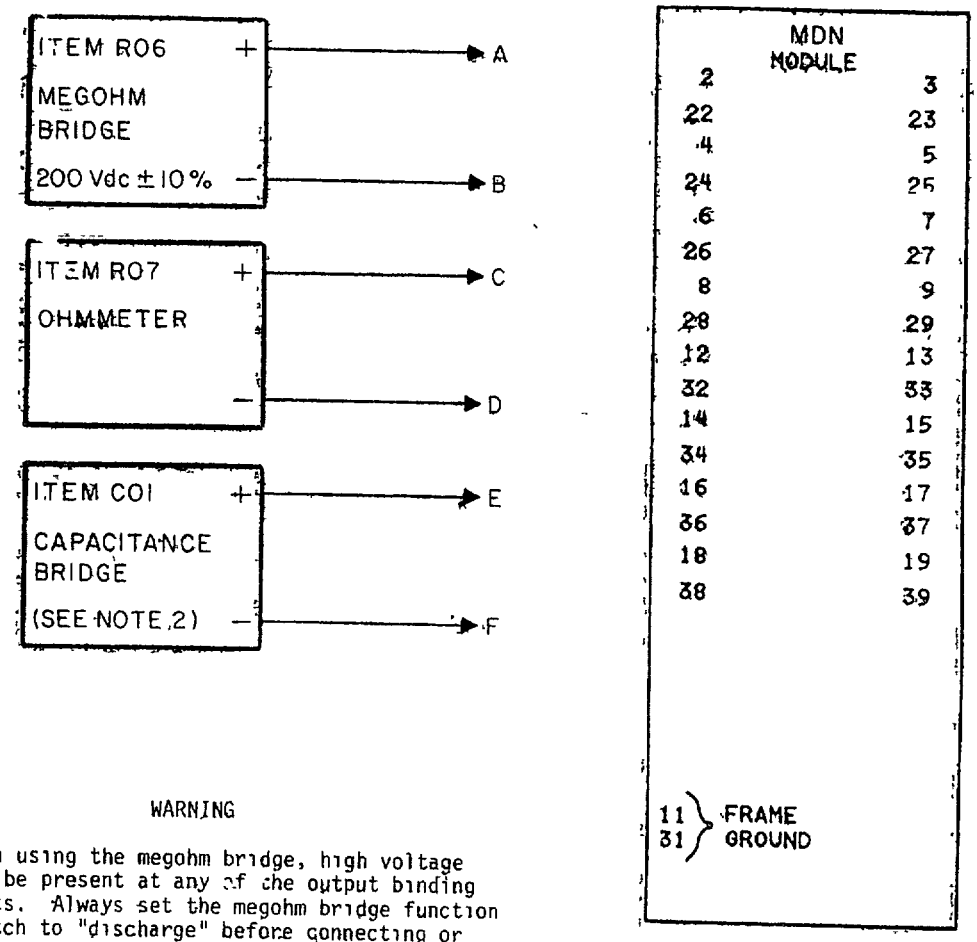
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
C1	R1 Resistance	2	3	73	77		3
C2	R9	22	23				
C3	R2	4	5				
C4	R10	24	25				
C5	R3	6	7				
C6	R11	26	27				
C7	R4	8	9				
C8	R12	28	29				
C9	R5	12	13				
C10	R13	32	33				
C11	R6	14	15				
C12	R14	34	35				
C13	R7	16	17				
C14	R15	36	37				
C15	R8	18	19				
C16	R16	38	39				

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TABLE V. Capacitance test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
D1	Capacitance	2	3	0	30	pF	3
D2		22	23				
D3		4	5				
D4		24	25				
D5		6	7				
D6		26	27				
D7		8	9				
D8		28	29				
D9		12	13				
D10		32	33				
D11		14	15				
D12		34	35				
D13		16	17				
D14		36	37				
D15		18	19				
D16		38	39				

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES

1. For test condition B2 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item R07.
2. For test conditions D1-D16

   a. All pins except those under test are interconnected.
   b. Test fixture and lead capacitance must be measured for each test condition prior to insertion of module into fixture and then subtracted from the capacitance measured with the module inserted. Test fixture and lead capacitance must not exceed 30 pF.
   c. Equivalent series capacitance shall be measured at a frequency of 10 kHz and a bridge voltage of 0.2 Vrms maximum.

FIGURE 3. Module integrity, continuity, resistance, and capacitance test circuit for tables II, III, IV, and V.

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TABLE VI Overshoot and undershoot test

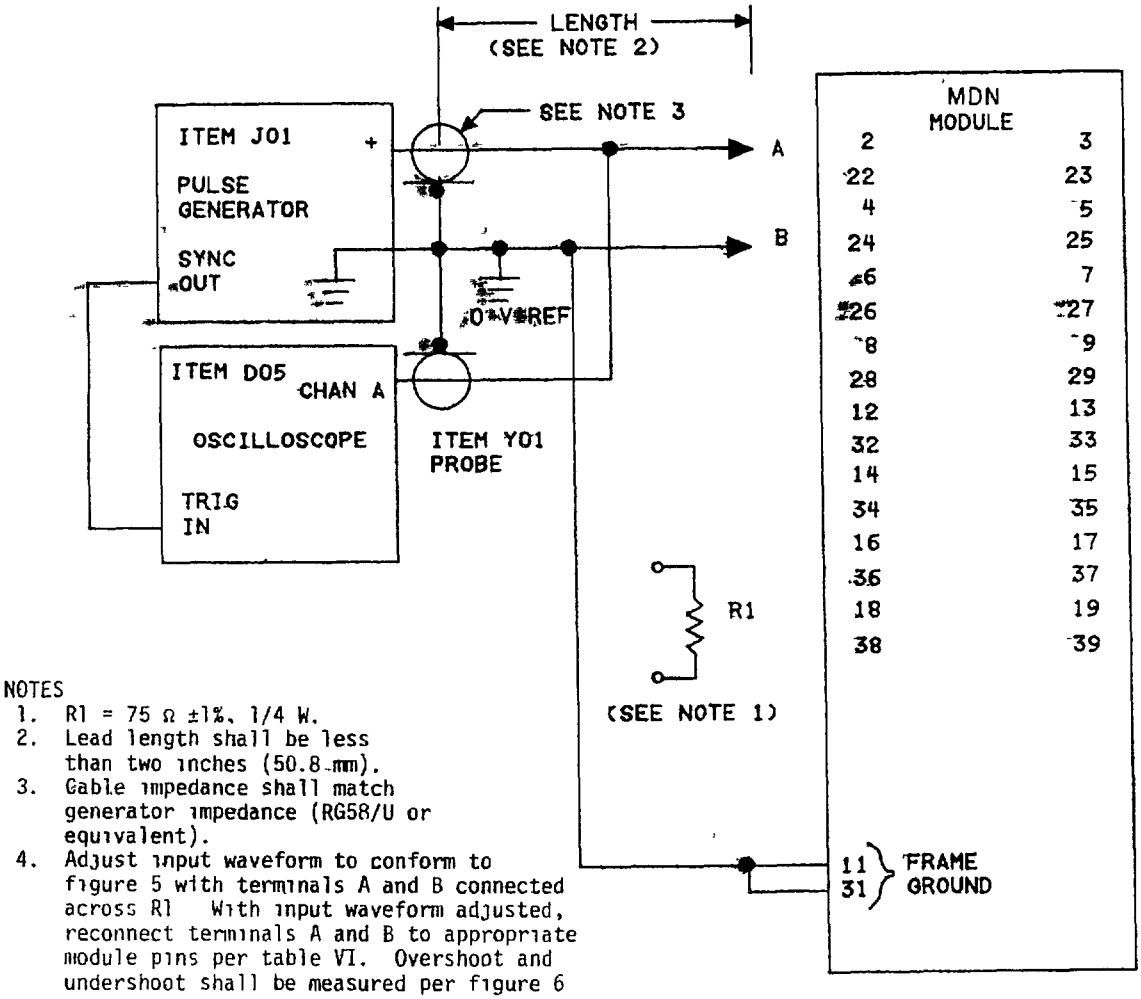
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	R1 Overshoot	2	3	0.0	0.3	V	4,5,6
E2	R9	22	23				
E3	R2	4	5				
E4	R10	24	25				
E5	R3	6	7				
E6	R11	26	27				
E7	R4	8	9				
E8	R12	28	29				
E9	R5	12	13				
E10	R13	32	33				
E11	R6	14	15				
E12	R14	34	35				
E13	R7	16	17				
E14	R15	36	37				
E15	R8	18	19				
E16	R16	38	39				
E17	R1 Undershoot	2	3				
E18	R9 Undershoot	22	23				
E19	R2 Undershoot	4	5				

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TABLE VI. Overshoot and undershoot test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E20	R10 Undershoot	24	25	0.0	0.3	V	4,5,6
E21	R3	6	7				
E22	R11	26	27				
E23	R4	8	9				
E24	R12	28	29				
E25	R5	12	13				
E26	R13	32	33				
E27	R6	14	15				
E28	R14	34	35				
E29	R7	16	17				
E30	R15	36	37				
E31	R8	18	19				
E32	R16	38	39				

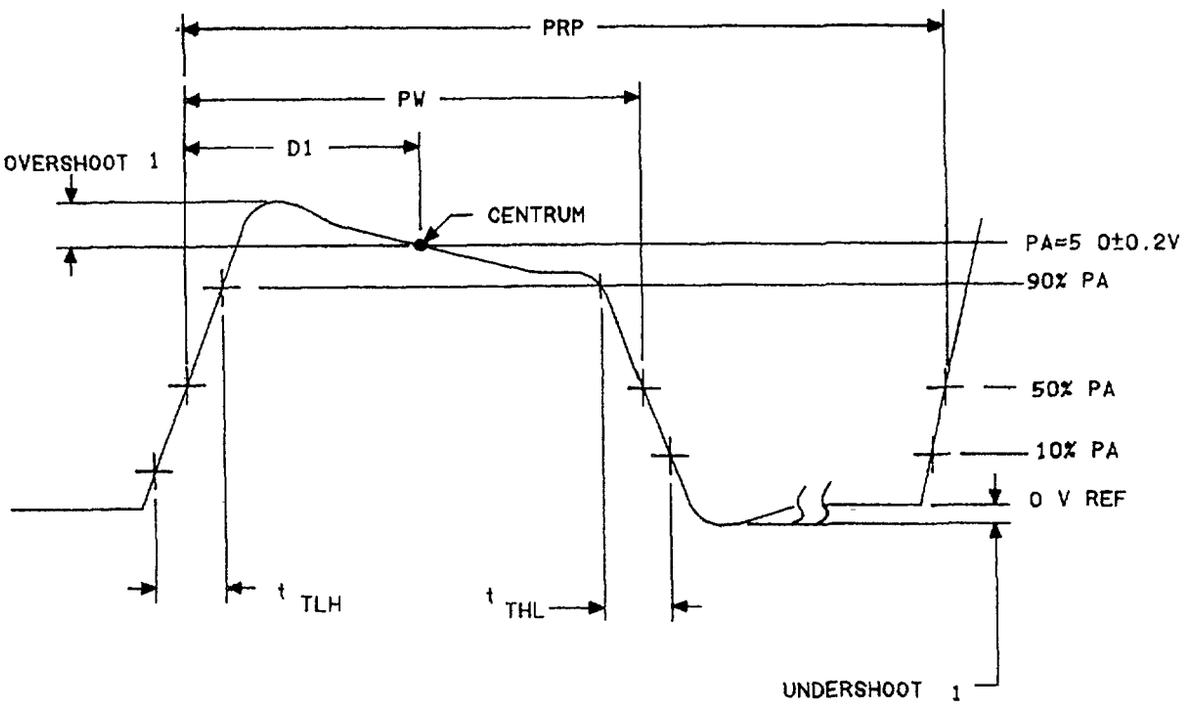
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NOTES
1. R1 = 75 Ω ±1%, 1/4 W.
2. Lead length shall be less than two inches (50.8 mm).
3. Cable impedance shall match generator impedance (RG58/U or equivalent).
4. Adjust input waveform to conform to figure 5 with terminals A and B connected across R1. With input waveform adjusted, reconnect terminals A and B to appropriate module pins per table VI. Overshoot and undershoot shall be measured per figure 6

FIGURE 4. Overshoot and undershoot test circuit for table VI.

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NOTE: Input pulse parameters

PW = 2.2 ±0.2 µs.

PRP = 11 ±1 µs.

$t_{TLH} = t_{THL}$ = 20 ±4 ns.

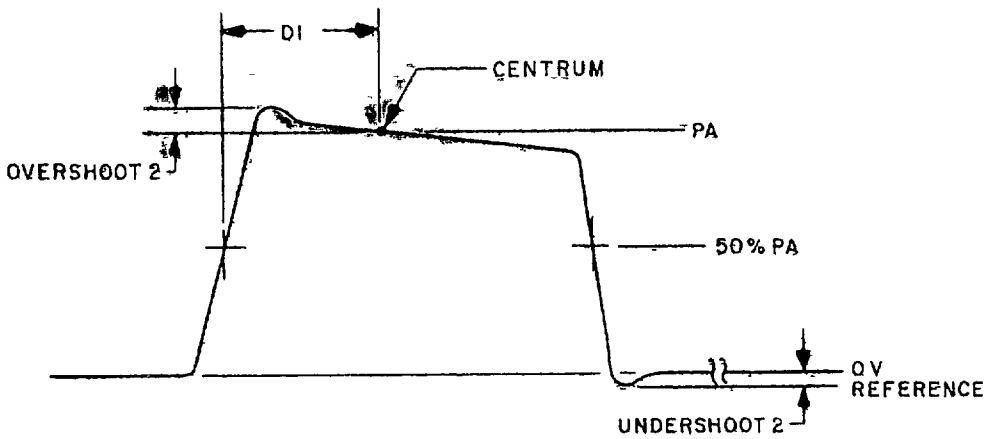
D1 = 1.2 ±0.2 µs.

Overshoot 1 = 0.25 V maximum

Undershoot 1 = 0.25 V maximum.

FIGURE 5 Overshoot 1 and undershoot 1 test circuit waveforms for table VI

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NOTES 1. D1 1.20 ±0.2 µs.

2. Overshoot and undershoot shall meet the applicable requirements of test condition E.

Overshoot = Overshoot 2 - Overshoot 1.
Undershoot = Undershoot 2 - Undershoot 1

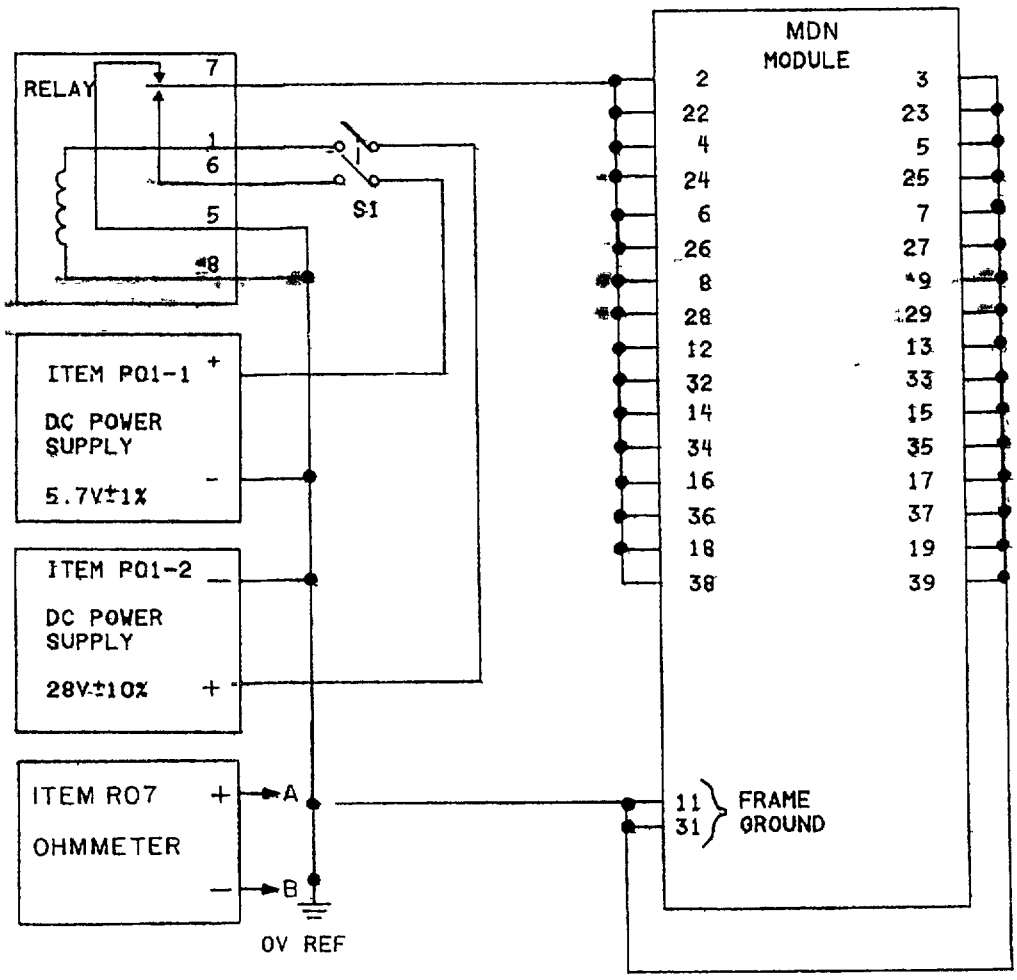
FIGURE 6. Overshoot 2 and undershoot 2 test circuit waveform for table VI.

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TABLE VII. Resistance surge test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	R1 Resistance surge	2	3	73	77	Ω	7
F2	R9	22	23				
F3	R2	4	5				
F4	R10	24	25				
F5	R3	6	7				
F6	R11	26	27				
F7	R4	8	9				
F8	R12	28	29				
F9	R5	12	13				
F10	R13	32	33				
F11	R6	14	15				
F12	R14	34	35				
F13	R7	16	17				
F14	R15	36	37				
F15	R8	18	19				
F16	R15	38	39				

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NOTES –

1. Relay = Agastat model 2112 or equivalent.
2. The relay shall be adjusted for a 1.0 ±0.2 second delay.
3. S1 = DPST switch.
4. To perform test, close switch S1 for a minimum of five seconds then open switch S1, connect module and measure the resistance as specified in table VII
5. Disconnect all pins from test setup prior to making measurements for test conditions F1-F16 Reconnect pins as required per table VII

FIGURE 7. Resistance surge test circuit for table VII.

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